

HEALTHCARE DATA VISUALIZATION, COST RELATIONSHIP & POLICY INSIGHTS

HEALTHCARE DATASET REPORT - WEEK 6

INTRODUCTION

The preprocessed Healthcare Dataset from Week 5 was analyzed using **Matplotlib and Seaborn** to understand patient demographics, billing patterns, insurance variations, admission trends, and relationships between healthcare-related variables.

OBJECTIVES

- Analyze patient age and gender distributions across medical conditions.
- Compare billing amounts across medical conditions, insurance providers, and admission types.
- Identify monthly patient admission trends.
- Analyze the relationship between age, hospital stay duration, and billing amount.
- Derive useful business insights for healthcare management.

KEY FINDINGS AND BUSINESS INSIGHTS

Analysis	Key Finding	Business Insight
Age & Medical Condition	Patient age distribution varies across medical conditions.	Demographic patterns can support better planning of healthcare services and resources.
Gender & Medical Condition	Gender distribution differs across medical conditions.	Understanding demographic composition can help hospitals plan patient-specific services and resource allocation.
Billing & Medical Condition	Billing amounts show variation across different medical conditions.	Management can monitor high-cost conditions and review associated resource utilization.
Billing & Insurance Provider	Billing amount distributions vary across insurance providers.	Comparing billing patterns can support financial monitoring and insurance-related cost analysis.
Billing & Admission Type	Emergency, Urgent, and Elective admissions show differences in billing patterns.	Admission-wise cost analysis can support budgeting and operational planning.

Monthly Admissions	Patient admissions vary across different months, with some months showing higher demand.	Monthly trends can help hospitals plan beds, staffing, and other resources during periods of increased demand.
Age, Stay Duration & Billing	Correlation analysis shows the relationships between age, hospital stay duration, and billing amount.	Monitoring stay duration and billing together can support cost and resource utilization analysis.

CONCLUSION

The analysis converts healthcare records into meaningful visual insights about patient demographics, costs, insurance patterns, and admission trends. These findings can support hospital management in resource planning, cost monitoring, capacity management, and data-driven operational decision-making.